Replacing the independent-observation likelihood

A proposal for the HIKET calibration

Revised 18 August 2026 — *agreed plan, not yet implemented*

> **In one paragraph**
>
> The calibration currently assumes that every soil-carbon measurement is an independent piece of evidence. It is not. The same plot measured in three campaigns gives three strongly related numbers; plots near each other give related numbers too; and every plot in one campaign shares whatever that campaign's method did. Counting 1205 measurements as 1205 independent facts overstates the evidence by roughly a factor of five for the quantity that controls the headline results. This document sets out the agreed plan: three shared-offset terms, all with their sizes fixed from outside the calibration, replacing the independence assumption. It gives the numbers, the traps found while deriving them, what the change is expected to achieve, and — equally important — what it cannot.

1 What the calibration does now

For plot i in campaign j the model returns a predicted carbon stock $f_{ij}(\theta)$, and the comparison with the observation is made on the log scale:

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2), \quad \sigma = 0.800.$$

The log-likelihood is the plain sum of the individual log-densities. That sum *is* the assumption that every e_{ij} is independent of every other. The assumption has never been stated as a hypothesis anywhere in the code or the manuscript; it arrived as a default.

2 Why that assumption is wrong here

Three different groupings make observations share information. They are easy to conflate, so we name them explicitly and use one notation throughout: a shared offset is u with a superscript naming what it groups.

term	groups	members	meaning
u_r^R	plots in a region	$\sim$150	same climate, parent material, forest history
u_i^P	visits to one plot	2.6	same soil, missing edaphic control
u_j^C	all plots in one campaign	$\sim$400	shared method, crew, laboratory

> **The counter-intuitive part**
>
> A *small* correlation shared by *many* observations does more damage to a national mean than a large one shared by few. The plot term has a correlation of 0.5–0.6 but only 2.6 members. The regional term has a correlation of just **0.080** — negligible-sounding — but $\sim$150 plots share it, and $1 + 149 \times 0.08 \approx 13$. Group size multiplies the effect. This is why the regional term matters more than the plot term, which is the opposite of what the raw correlations suggest.

2.1 Measured, from the observations rather than from residuals

> **Design principle**
>
> Residuals may **check** a design value. They may never **set** one. Setting the likelihood from the residuals of a fit produced under the assumption we are replacing is circular. Every number below comes from the observations themselves, with campaign levels removed first so nothing can be a trend artefact.

The plot term. Under compound symmetry the covariance between any two visits to the same plot *is* τ_P^2, so it can be read directly off a campaign pair with no variance decomposition and no distributional assumption:

pair	n	correlation	τ_P
1985–2006	384	0.805	0.355
1985–2024	342	0.762	0.363
2006–2024	**442**	**0.820**	**0.396**

We adopt $\tau_P = 0.396$ from the 2006–2024 pair: the largest sample, the widest of the three, and **the only pair untouched by the 1985 problems** — no imputed litter layer, no 1986–1995 dating spread. The 1985 campaign therefore enters the likelihood at full weight but sets none of the likelihood's parameters.

The regional term. Splitting the plots into eight equal-count latitude bands (ETRS northing), after removing campaign levels, the band means run monotonically south to north:

$$+0.18,\ +0.16,\ +0.06,\ +0.01,\ -0.03,\ -0.16,\ -0.12,\ -0.10$$

a gradient of about 32%. Between-band sd = **0.117**, between-plot within band = 0.397, giving a regional correlation of 0.080. We adopt $\tau_R = 0.117$.

The campaign term. Not measurable from this data at all — see Section **??**.

2.2 There is no field replication to check any of this

The Biosoil workbook carries a `REP` column, but only 63 rows of 3028 (about 16 plots) have `REP` = 2, all in Biosoil, none in 1985 or 2024. Of the three carbon columns, two return sd(log ratio) = 0.000 and 0.003 between the pairs — copied values, not independent re-measurements. Only one shows genuine variation, at about 12% per sample, and that reads as a laboratory duplicate rather than a second field sample.

> **Consequence**
>
> We cannot empirically separate "we sampled a different spot" from "this plot really is different". Random relocation between campaigns is harmless for the design — it is genuinely independent between visits, so it already sits inside σ_e — but a *systematic* component of relocation or protocol drift is a campaign-level offset, and nothing in the data can measure it.

3 Why simply assuming a larger error does not work

Inflating σ multiplies every direction of the posterior by $\sqrt{k}$. Reaching what is needed would require $\sigma \approx 1.8$ against a measured residual spread of 0.71, which would destroy the self-consistency that justifies σ in the first place. **Correlation widens the level at fixed total variance; nothing else available does.** That is the whole argument for this instrument, and it is why the variance is *split* rather than *added*:

$$\sigma^2_{\text{total}} \;=\; \tau_R^2 + \tau_P^2 + \tau_C^2 + \sigma_e^2, \qquad \sigma_{\text{total}} = 0.800 \text{ (unchanged)}.$$

A trap found while deriving this — do not repeat it

The observations show an intraclass correlation of **0.799**. Transplanting that *ratio* into a total of 0.800 gives $\tau = 0.715$ and $\sigma_e = 0.359$ — *below* the models' own within-plot error of 0.44–0.48. That would assert the trend is measured more precisely than any model can reproduce it: the same overconfidence we are removing from the level, recreated in the trend. The 0.799 is a ratio inside the observations' own spread of 0.422; only its *absolute* part transfers. Always carry absolute standard deviations across, never ratios.

4 The plan

4.1 The model

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; u^R_{r(i)} \;+\; u^P_i \;+\; u^C_j \;+\; e_{ij}$$

with $u^R \sim N(0, \tau_R^2)$, $u^P \sim N(0, \tau_P^2)$, $u^C \sim N(0, \tau_C^2)$, $e \sim N(0, \sigma_e^2)$.

4.2 Nothing new is estimated

None of the offsets is ever given a value. They are **marginalised** — integrated out, averaged over all the values they might take — leaving behind only their effect on the covariance:

$$\Sigma \;=\; \sigma_e^2 I \;+\; \tau_P^2 Z_P Z_P' \;+\; \tau_R^2 Z_R Z_R' \;+\; \tau_C^2 Z_C Z_C'.$$

This distinction is the crux of the whole proposal and is worth stating plainly:

	what happens to the offset	consequence
ignore it	fixed at exactly zero	asserts perfect comparability
fit it	data choose the best-fitting value	**eats the trend** ($\delta = -0.137$)
marginalise it	never given any value	widens the right direction

The standing rule that "a campaign-specific bias is forbidden" applies to the *fitted* version, which is unidentifiable and absorbs the signal. The marginalised version adds no parameter and cannot absorb anything, because there is nothing to absorb *with*.

4.3 The numbers

term	value	source	status
τ_R	0.117	latitude-band means, campaign-centred	measured, fixed
τ_P	0.396	2006–2024 pair covariance	measured, fixed
τ_C	0.06 (1985), 0.03 (2006, 2024)	prescribed, see below	**assumed**
σ_e	0.685	remainder of the fixed total	derived

The one-sided check

$\sigma_e = 0.685$ sits comfortably above the models' measured within-plot error of 0.44–0.48, so the trend is not over-claimed. This is a residual-based *check*, not a residual-based *setting* — the distinction the design principle allows.

4.4 Setting τ_C

τ_C cannot be measured; it is chosen. That is not a weakness relative to the status quo, because the status quo also chooses a value — zero — and zero is the least defensible number available, asserting that three campaigns spanning 39 years, two laboratory methods and three field crews are perfectly inter-comparable.

The choice rests on two arguments that happen to agree. First, the largest campaign-level defect we have actually identified and corrected — the 1985 litter-layer definition — was worth 3.37–4.27 $\mathrm{Mg\,C\,ha^{-1}}$, i.e. **4.8–6.0% of stock**, and at least three further mechanisms are documented but unquantified (LOI-derived mineral carbon in 1985 versus dry combustion later, relocated subplots, the 1986–1995 dating spread). The residual should be *at least* the size of the one we happened to measure. Second, 1985 deserves roughly twice the others: different laboratory method, relocated subplots, an imputed litter layer, a ten-year dating spread, and no external validation, whereas the 2006 and 2024 processing reproduces LUKE's official national stocks exactly. A 3% base doubled for 1985 gives 6%, which is also the top of the measured litter-layer range. **This is a stated assumption, not a measurement, and should be written as one.**

5 What the change is expected to achieve

5.1 Effect on the evidence

On the calibration target (1205 observations, 456 plots; 310 with three visits, 129 with two, 17 with one):

structure	SE of the national mean	inflation	effective n
independent (now)	0.0231	1.00	1205
$+\,u^P$	0.0274	1.19	855
$+\,u^P + u^R$	0.0496	**2.15**	**260**

> **The plot term alone is not worth doing**
>
> u^P by itself reduces precision by a factor 1.4, against a likelihood that is already 10–15 times more informative than the prior for σ_{input}. It moves the estimates by 1–2%. **The regional term is where the effect is**, and the two must go in together.

5.2 Expected landing

Marginal, local approximations — direction is predictable, magnitude is not; an estimate of exactly this kind gave an effective sample size of 1 in 1001 when used to predict a prior change. Treat as order of magnitude.

	now	expected
posterior sd of $\log \sigma_{\text{input}}$	0.062–0.076	roughly double
σ_{input}	1.69–2.10	1.51–1.80
intrinsic MRT (Yasso07/15/20)	21.7 / 25.1 / 21.4	∼25.4 / 28.5 / 24.8
effective litter flux	4.25–5.28	3.8–4.6

The flux row is the checkable one: all six models would fall inside the Nordic Class I window of 4.62 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ for the first time.

No prediction is offered for σ_{init}

The arithmetic above assumes the likelihood loosens uniformly in every direction. That is apt for a level parameter such as σ_{input} and *wrong* for σ_{init}, whose information lives in trajectory shape — and the shared offsets cancel on differencing, leaving within-plot contrasts untouched. σ_{init} should move only as a knock-on through its correlation with σ_{input}, measured at -0.29 to -0.45. An earlier reading of the marginal prior-to-posterior widths concluded the opposite and was wrong.

6 What it will not achieve

6.1 The trend is a separate problem, and the short interval is the fragile one

The correlated likelihood widens the *level*. It does nothing for the *trend*, which is governed by campaign comparability. Writing the implied uncertainty on an observed rate as $\tau_C\sqrt{2}\,\bar{C}/\Delta t$:

	signal	% of stock	τ_C that dissolves it (1σ)
1985→2024	9.1 Mg ha^{-1} over 35.1 yr	12.8	**9.1%**
2006→2024	2.1 Mg ha^{-1} over 18.0 yr	**3.0**	**2.1%**

τ_C scheme (1985 / others)	1985–2024 (+0.259)	2006–2024 (+0.117)
0 (current, implicit)	exact	exact
4% / 2%	±0.090 (2.87σ)	±0.111 (1.05σ)
6% / 3% (adopted)	±0.135 (1.91σ)	±0.167 (0.70σ)
10% / 5%	±0.226 (1.15σ)	±0.278 (0.42σ)

The main result of this analysis, and it is not about the likelihood

The intuition has been that 1985 is the campaign to distrust, so the long comparison is the weak one. The arithmetic says the reverse. **1985–2024 is robust** — 35 years is long enough for the signal to outgrow any plausible bias, surviving to $\tau_C \approx 9\%$. **2006–2024 is not resolvable** at any τ_C above about 2%, even though it uses our two best campaigns: same protocol, both reproducing LUKE's official stocks exactly. An 18-year change of 3% of stock simply cannot survive a campaign offset of comparable size. No amount of trusting those two campaigns repairs this; the interval is too short relative to the signal.

A recorded result that must be restated

The finding that "2006→2024 is a source in all six models" has died, with four models now bracketing the observed +0.117. That is much weaker than it reads: +0.117 itself is not resolvable against any campaign bias above ∼2%. "The models bracket the observation" is not evidence when the observation carries an unquantified systematic of its own size.

This table is analytic — arithmetic on the campaign means — so all three τ_C schemes are reported at no computational cost, whatever is run.

6.2 Other things it does not fix

The plot term is not a substitute for edaphic predictors; no model in the ensemble has any, which is why $R^2 \approx 0.02$ and why a persistent per-plot offset is the right shape for the missing information. And nothing here addresses the possibility that the models are structurally wrong about early accumulation — the misfit concentrated in 1985–2006 remains whatever the likelihood says.

7 Implementation

7.1 It is computationally trivial, for one specific reason

Every variance component is **fixed**. Σ therefore does not depend on θ and does not change between MCMC draws. Factorise it *once* at setup — a 1205×1205 Cholesky, well under a second, about 12 MB stored — and each likelihood evaluation is a triangular solve for $r'\Sigma^{-1}r$ with $r = \log y - \log f(\theta)$. That is of order 1 ms against a forward model costing 227 ms: under half a percent of overhead. $\log |\Sigma|$ is computed once.

Woodbury and Sherman–Morrison identities are *not* required. They would earn their place only if the variance components were being estimated, or if n were far larger.

> **The one structural change**
>
> The likelihood is currently computed plot-by-plot under `mclapply` and summed. The plot term preserves that decomposition and the regional term preserves it with regions as blocks, but **the campaign term couples every plot and every region simultaneously**, so the per-plot sum is replaced by a single global solve. This is a change to how the likelihood is assembled, not a local edit.

7.2 Build order — these are local unit tests, not calibrations

Each check is a single likelihood evaluation at a fixed parameter vector, compared against the previous stage. Seconds on a laptop; no MCMC.

1. $\tau_P = \tau_R = \tau_C = 0$ must reproduce the current log-likelihood **exactly**, to floating point. This is the only test that separates a correct implementation from a plausible-looking one.
2. Add u^P; setting $\tau_P = 0$ must return to step 1.
3. Add u^R; setting $\tau_R = 0$ must return to step 2.
4. Add u^C; setting $\tau_C = 0$ must return to step 3.

7.3 Run configuration

switch	value	note
correlation	$u^R + u^P + u^C$	values in the table above
`HIKET_SIGMA_TOTAL`	0.800	unchanged; split, not added
`HIKET_LIK_DF`	**unset**	Student-t dropped, see below
`HIKET_SIGMA_1985_INFL`	**1.0**	replaced by $\tau_C(1985)$=0.06
chains / iterations	5×50000	unchanged

One launch, six jobs, all six models. No staged runs: the intermediate stages are unit tests, not calibrations.

Why Student-*t* is dropped. The tails were adopted because measured kurtosis is 6.9–7.1 against a Gaussian 3. But pooling observations from plots with different persistent offsets produces heavy tails *mechanically* — a mixture of Gaussians with different means is leptokurtic. Tails and shared offsets are competing explanations for the same measurement, and fitting both risks paying twice for one structure. Drop them, then measure kurtosis *conditional on* the fitted offsets: if it remains near 7 the tails are independently justified and return; if it falls toward 3 they were a proxy for exactly this. Note that σ means the same thing either way, since the current setup already maps it to the standard deviation.

The 1985 inflation is replaced, not removed. 1985 keeps its extra weight — $\tau_C = 0.06$ against 0.03 for the others, the factor of two argued for above. What is switched off is the old *device*. `SIGMA_1985_INFL` $= 2.0$ inflated the *independent, per-observation* noise for 1985, reaching the campaign level only because n is finite (as $1/\sqrt{n}$) and degrading every plot-level comparison in 1985 on the way. τ_C is a *shared* level offset: it displaces the campaign level and does nothing else, whatever n is. Since the documented defect is a level and not scatter — 1985's noise level is normal, subsoil repeatability 0.63 against 0.65 — the old device was a blunt proxy that happened to hit the right target, and τ_C hits it directly. It also does what the old device structurally cannot: express level uncertainty for 2006 and 2024, which is what limits the recent trend.

Numerically the two are near-equivalent for 1985, which is exactly why they must not be stacked. On the 1985 campaign mean, the other terms alone give 0.0578; the old device adds 0.0605 in quadrature and τ_C adds 0.0600. **Keeping both gives an effective τ_C of 0.085** — silently the 10%/5% broad scheme rather than the 6%/3% one adopted.

> **This run changes three things at once**
>
> Correlation on, tails off, the 1985 inflation replaced by τ_C. It is therefore a comparison of *error models*, not an isolation of the correlation effect, and the one-factor discipline used in previous runs does not apply. Accepted deliberately: the intermediate configurations are not scientifically interesting, and attribution can be recovered afterwards from the conditional kurtosis and the campaign-specific residual spreads.

8 Pre-registered predictions

Stated before launch so that the outcome cannot be rationalised afterwards.

1. σ_{input} falls; MRT rises.
2. Posterior widths increase, most along the level direction.
3. RMSE is unchanged or slightly worse; the calibration–holdout gap narrows or holds.
4. σ_{init} moves only weakly, and only via its correlation with σ_{input}.

Falsifier: if posterior widths do *not* increase, the implementation is not doing what the theory says, whatever the medians do.

> **Comparison rules**
>
> Log-likelihoods are **not** comparable across this change — the normalising constant moves. Compare on the RMSE distributions (`S14_rmse_posterior`). And the standing rule applies with full force: **τ_C, τ_P and τ_R must never be tuned to land MRT on a published value.**

9 What this is adopted for

Not because it moves MRT. The independence assumption is measurably false: 1205 observations carry the weight of about 260 for the national level. The corrected accounting is the more defensible error model whether or not the estimates move at all. If MRT does not move, that is a result — it would say the short residence times are not an artefact of overconfidence, which is worth as much as the alternative.